Audit of the SV efficiency proof, round 7: the article version

Source audited: the article `sv_efficiency_proof.tex` with `numeric.tex`, at commit "SV numerics: 3D results, summary, tables next to their discussion". The article renumbers the working note `sv_efficiency_proof.md`; its header maps the numbering. Earlier rounds: 1–6. Not refereed.

Part A. Proof-check

A1. Scope

This round checks:

- the parts that are new in the article: Remark 2.1 (edge form ⇒ (H6)), the proof of Lemma 2.2 (κ), the restated estimator of §3 (η = η_A + η_N + η_S, η_E = κη + η_S), the hypotheses of Theorem 4.2, the proofs of Corollaries 4.3 and 4.4, Lemma 5.4, Lemma 5.11(iii), Remark 7.3 (U1 fix) and Proposition 8.1 with the new split η_disc = η_E − η_alg;
- the consistency of the code with these statements (stopping split, κ);
- the claims of §10 against the data in `examples/convection_diffusion/sv_numerics/output/*.json`.

A2. Step table

tep	tool	hypothesis	constant	status	
em. 2.1	$u(x_j)-u(x_i)=\nabla u	_K\cdot t_E$ for P_1; $\sum_i\sum_j\beta_{ij}(u_j-u_i)v_i=\sum_E\beta_E(u_j-u_i)(v_j-v_i)$	$\beta_{ij}=\beta_{ji}\ge 0$	exact	OK. Rows of Dirichlet nodes drop out since $v_i=0$. Non-symmetric limiters (Kuzmin's) are excluded, as stated
emma 2.2	$B(v,v)=\varepsilon\|\nabla v\|^2+((c-\frac12\mathrm{div}\,b)v,v)+\frac12\int_{\Gamma_N}b\cdot n\,v^2$; $B=B_S'+B_A'$	(H3)–(H5)	$\kappa=2+\max\{1,c_b\}$	OK. $B=B_S'+B_A'$ holds **exactly**; only the skew-symmetry of B_A' is "up to boundary terms" (**V4**, wording)	
3, Thm 4.1	$E=\tnorm{e}_\oplus+\|\tilde S\|_*\le\kappa\eta+\eta_S$	—	exact	OK; matches the code (`eta_E = κ η_int + η_S + aug_bd`, with η_int = min{η_dir, η_A + η_S}, Remark 3.2)	
hm 4.2, hypotheses	—	lists (H1), (H2), (H4)–(H6)	—	OK but not minimal: the inflow part of (H4) and $b\in W^{1,\infty}$ are not used; $b,c\in L^\infty$ suffice (**M1**)	
or. 4.3	$\eta_E\le(\kappa+1)\eta$, $\eta\le C\max\{1,c_b\}(E+\mathrm{osc})$	—	$C(1+c_b)^2$	OK	
or. 4.4	$\eta_{E,c}=(M-1)\eta+\eta_E$	—	text: $M-1+\kappa=2M+1$	**V2**: $\eta_E\le(\kappa+1)\eta$, so the factor is $M+\kappa=2M+2\le4(1+c_b)$; still linear	
emma 5.4	$\sum_{a\in\mathcal V_K}\nabla\psi_a=0$; Friedrichs on patches with a free face	—	$C_I(d,\vartheta)$	OK (re-derived)	
emma 5.10, reaction egime	optimality $J_a(\sigma^a)\le J_a(v_b)$; $\rho\le\|\tau^a\|$	broken RTN_1 data	$C(d,\vartheta)$	OK	
emma 5.11(iii)	(ii)	—	—	**V3**: "at an iterate, bounded by $C\|\mathrm{Res}_A\|_*$" holds for the global term only under Assumption 5.5; say so	

tep	tool	hypothesis	constant	status	
em. 7.3	Friedrichs on H^1_D: $(\sum_a\mu_a^2)^{1/2}\le\sqrt{d+1}\,C_PC_F\|c-\frac12\mathrm{div}\,b\|_\infty h_{\max}\varepsilon^{-1}\tnorm{e}$	σ = 0	—	OK (U1 closed)	
rop. 8.1	$\eta_E=\eta_{\rm disc}+\eta_{\rm alg}$ by definition	—	—	OK. The code now uses the same split (`EstimatorStop._sv`, commit a56597e), and `test_adaptive_stop` checks the identity	
in the code	$c_b=\|c-\frac12\mathrm{div}\,b\|_\infty/\sigma$	the sup over Ω	—	**V1**: the code samples the sup (see A4)	
10 claims	the JSON data	—	—	OK: no run has I, $I_\oplus$, I_S, I_E or $I_{E,c}<1$; the quoted ranges match the data after the round-7 corrections (I_E 2.6–5.3 in 2D for constant coefficients; 6.0–7.1 at σ = 1, ε ≤ 10^{-7})	

A3. Verdict

- **The mathematics is correct**, given A2 (Neumann version, [EV15]) and A5 (Licht 2019, abstract-level).
- **V2–V4** are wording or bookkeeping; none changes a statement.
- **V1** is an implementation gap that affects the claim "guaranteed" in practice, not the theorem.

A4. V1: the computed κ is not a guaranteed upper bound for variable c

- **What the theorem needs.** $c_b\ge|c-\frac12\mathrm{div}\,b|_\infty/\sigma$ (or the Friedrichs entry), so that κ is an upper bound of the constant in Lemma 2.2.
- **What the code computes.** `flux_estimator.py` takes `cmb` as the maximum of $|c-\frac12\mathrm{div}\,b|$ at the interior quadrature points, with $\mathrm{div}\,b$ by central differences. This is a lower estimate of the supremum whenever the maximum lies on a cell boundary.
- **Observed.** For $c=1+(c_{\max}-1)xy$ the maximum is at the vertex $(1,1)$; the code reports $c_b=981.4$ ($n=8$), 990.7 ($n=16$) and 995.3 ($n=32$) for $c_{\max}=1000$. κ is therefore about 2 % too small on the coarsest mesh. In these runs $E\le\eta_E$ still held, by a wide margin, but the bound is not certified.
- **Also unchecked:** the hypothesis $c-\frac12\mathrm{div}\,b\ge\sigma$ itself; σ is supplied by the user and not compared with the data.
- **Fix (code only).** Bound $|c-\frac12\mathrm{div}\,b|$ cellwise from above, as `eps_cell_bounds` does for ε:
  - exactly for piecewise P_1 data, from the vertex values;
  - otherwise by the sampled maximum plus h_K times a sampled gradient bound, flagged in `info` as not guaranteed;
  - optionally accept a user-supplied bound `cb_bound`.

Check $\min(c-\frac12\mathrm{div}\,b)\ge\sigma$ on the same samples and warn if it fails. The article needs no change beyond one sentence in Remark 2.3.

Status of the fixes (19 Oct 2026)

- **V1:** fixed in the code.
  - `flux_estimator.reaction_cell_bounds` samples $g=c-\frac12\mathrm{div}\,b$ at the quadrature points, the vertices and the edge midpoints of every cell. It returns the exact vertex extrema where g is affine, and otherwise the sampled extrema widened by $\frac{h_K}2\max|\nabla g|$; those cells are reported in `info`.
  - c_b and κ use this bound.

  - $g\ge\sigma$ is checked on the samples, with a warning if it fails.
  - For $c=1+999xy$ the computed c_b is now 1125, 1062 and 1031 ($n=8,16,32$; true value 1000) instead of 981–995.
  - The new test `test_reaction_cell_bounds` checks: exact for affine c, above the true supremum for $c=1+999xy$, and the σ warning. All seven estimator test files pass.
  - The article states the procedure in Remark 2.3. The variable-reaction experiment was re-run, and Table 3 and §10.4 are updated.
- **V2:** fixed. In the proof of Corollary 4.4, $\eta_E\le(\kappa+1)\eta$ gives the factor $M+\kappa=2M+2\le4(1+c_b)$.
- **V3:** fixed. Lemma 5.11(iii) now separates the local term ($\le C|\mathrm{Res}_A|_{,\omega_a}$), the true dual norm of the global term ($\le C_I|\mathrm{Res}_A|_$) and its computed bound ($\le C_\Pi C_I|\mathrm{Res}_A|_*$, Assumption 5.5).
- **V4:** fixed. §2 now says that $B=B_S'+B_A'$ is exact and that $B_A'(w,v)+B_A'(v,w)=\int_{\Gamma_N}b\cdot n\,wv$ on H^1_D.
- **M1:** not changed (not requested).

A5. Optimality

- **M1.** Theorem 4.2 can be stated under (H1), (H5), (H6), Γ_D of positive measure, and $b,c\in L^\infty$. The inflow condition and $b\in W^{1,\infty}$ enter only through Lemma 2.2, that is, through Corollaries 4.3–4.4.
- **The cellwise-σ track** (energy norm with the full $c-\frac12\mathrm{div}\,b$, in the spirit of [AV14]) remains the most useful extension. It would remove:
  - the c_b factors;
  - the restriction of Theorem 7.2;
  - the term μ of Remark 7.3;
  - and, with it, V1, because κ would no longer depend on a supremum of the data.

The numerics confirm the cost of the current track: η_E/E reaches $5.2\,c_b$ at $c_b\approx10^3$.

Part B. Literature scan (round 7)

B1. Answer

- **No change in novelty or track.** Searches for 2025–2026 work turned up nothing that comes closer to the article than what earlier rounds covered:
  - on equilibrated-flux estimators for algebraic flux correction;
  - on a posteriori stopping for algebraically stabilised schemes;
  - on robust equilibrated bounds with a variable reaction coefficient.
- **The closest work remains:**
  - for algebraic stabilisations: the residual-based estimators of Jha (2021, 2024) and the numerical study of Ahmed and Jha [AJ27];
  - for linear stabilisations, in the augmented norm: Eigel–Merdon (2016) and Cai & Cai (2025);
  - for reaction–diffusion: Smears–Vohralík (2020) and Ainsworth–Vejchodský (2014);
  - for stopping, in a monotone or diffusion setting: Ern–Vohralík (2013) and Haberl–Praetorius–Schimanko–Vohralík (2021).

B2. Frontier additions

result	scheme	estimate	relevance
A robust a posteriori error estimator for the Oseen problem, arXiv 2604.23453 (2026)	Oseen	robust (per the title)	the vector analogue of the convection setting; read before extending to Oseen [title only]
Equilibrated-flux estimates for nonlinear doubly degenerate elliptic problems, CAMWA 195 (2025)	conforming, nonlinear diffusion	guaranteed, robust in the degeneracy, locally efficient	context for nonlinear operators; no convection [search summary]

B3. Open items, ranked by novelty-to-effort

1. **V2–V4 and M1:** text only.
2. **V1:** certified cellwise bound of $c-\frac12\mathrm{div}\,b$ in the code. Low effort.
3. **The cellwise-σ track.** Medium effort; removes c_b, the Theorem 7.2 restriction, μ and V1.
4. **A5:** read Licht 2019, Theorem 1.
5. **The stopped iterate against the discrete solution.** Open, high risk.
6. **Oseen extension:** check arXiv 2604.23453 first.

B4. References (new in round 7)

- A robust a posteriori error estimator for the Oseen problem, arXiv 2604.23453. [title only]
- Equilibrated flux a posteriori error estimates and adaptivity for nonlinear and doubly degenerate elliptic problems, Comput. Math. Appl. 195 (2025). [search summary]
- Searched again: arXiv 2005.02938 (Jha 2021), arXiv 2107.06341 (Cai & Cai), arXiv 1812.06678 (SV20), arXiv 1401.2394 (AV14), arXiv 2004.13137 (HPSV21).